

Data Engineering Roadmap

1. Overview

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This roadmap is designed to build strong foundations in Data Engineering, covering core concepts, tools, and real-world project implementation.

Focus Areas

- Core Concepts
 - Programming and Data Processing
 - Cloud and Big Data Technologies
 - Real-Time Project Implementation
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2. Learning Roadmap

Phase 1: Foundations of Data Engineering

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2.1 Data Engineering Fundamentals

- What is Data Engineering
- Roles and responsibilities of a Data Engineer
- Data Engineer vs Data Analyst vs Data Scientist
- Data lifecycle:
 - Data ingestion
 - Data processing
 - Data storage
 - Data consumption
- Types of data:
 - Structured
 - Semi-structured
 - Unstructured

2.2 Data Pipeline Concepts

- ETL vs ELT
- Batch processing vs Real-time processing
- Data pipeline architecture

- Data flow design patterns
- Introduction to tools:
 - Apache Kafka
 - Apache Spark
 - Apache Airflow
- Data Lake vs Data Warehouse

2.3 Databases

SQL Databases

- RDBMS concepts
- ACID properties
- Schema design basics
- Normalization

NoSQL Databases

- Types:
 - Key-Value
 - Document
 - Column
 - Graph
- Use cases of SQL vs NoSQL

2.4 Linux Fundamentals

- File system structure:
 - /home, /etc, /var, /tmp
- File operations:
 - ls, cd, pwd, cp, mv, rm, touch
- File viewing:
 - cat, less, head, tail
- Searching and processing:
 - grep, find, awk, sed
- File permissions and ownership
- Process management:
 - ps, top, kill

- Job scheduling:
 - cron
- Shell scripting basics:
 - variables, loops, conditions

2.5 SQL Fundamentals

- SQL execution order
- Basic queries:
 - SELECT, WHERE, ORDER BY
- Aggregations:
 - GROUP BY, HAVING
- Joins:
 - INNER, LEFT, RIGHT, FULL
- Functions:
 - COUNT, SUM, AVG, MIN, MAX
- Data manipulation:
 - INSERT, UPDATE, DELETE
- Constraints:
 - PRIMARY KEY, FOREIGN KEY, UNIQUE
- Index basics

Phase 2: Programming and Data Processing

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3.1 Python Fundamentals

- Syntax and structure
- Data types:
 - int, float, string, boolean
- Data structures:
 - list, tuple, set, dictionary
- Control flow:

- if-else, loops
- Functions and modules
- Exception handling

3.2 Python for Data Engineering

- File handling:
 - CSV, JSON, TXT
- Data processing scripts
- Working with APIs (requests)
- Logging and error handling
- Virtual environments

3.3 Python Libraries

pandas

- DataFrames
- Filtering and selection
- Grouping and aggregation
- Merging and joining

numpy

- Arrays
- Numerical operations

3.4 PySpark Fundamentals

- Spark architecture
- RDD vs DataFrame vs Dataset
- SparkSession
- Transformations:
 - map, flatMap, filter
- Actions:
 - collect, count, show
- Reading and writing data:
 - CSV, JSON, Parquet

3.5 Advanced PySpark

- Joins and aggregations
- Window functions
- Partitioning:
 - repartition, coalesce
- Performance optimization:
 - caching, broadcasting

Phase 3: Cloud and Data Engineering Tools

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4.1 Cloud Fundamentals

- Cloud computing concepts:
 - IaaS, PaaS, SaaS
- Benefits of cloud computing

4.2 Microsoft Azure

- Azure Blob Storage
- Azure Data Lake
- Azure SQL Database
- Azure Databricks
- Azure Data Factory
- Resource management:
 - Azure Portal
 - Azure CLI

Phase 4: Advanced Data Engineering

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5.1 Advanced SQL

- Subqueries (correlated and non-correlated)
- Common Table Expressions (CTEs)
- Window functions:
 - ROW_NUMBER, RANK, DENSE_RANK

- LEAD, LAG
- Case statements
- Query optimization
- Indexing strategies
- Partitioning

5.2 Data Warehousing

- OLTP vs OLAP
- Star schema
- Snowflake schema
- Fact and dimension tables
- Slowly Changing Dimensions (SCD)
- Data marts

5.3 Data Pipelines

- ETL pipeline design
- Data ingestion:
 - APIs
 - Files
 - Databases
- Data transformation
- Data validation

5.4 Workflow Orchestration

- Apache Airflow:
 - DAGs
 - Task scheduling
- Azure Data Factory:
 - Pipelines
 - Linked services

5.5 Best Practices

- Data quality checks
- Handling missing data
- Logging and monitoring

- Version control using Git
- Testing:
 - Unit testing
 - Integration testing

Phase 5: Real-Time Project

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6.1 Project Scope

- Data ingestion:
 - APIs
 - Databases
 - Files
- Data transformation:
 - Python
 - PySpark
- Pipeline development:
 - Batch or real-time
- Data storage:
 - Azure Blob
 - Data Lake

6.2 Advanced Components

- Streaming (optional):
 - Apache Kafka
 - Spark Streaming

6.3 Automation

- Apache Airflow
- Azure Data Factory

6.4 Deployment

- Cloud deployment
- Monitoring and logging
- Performance optimization

6.5 Deliverables

- Source code (Git repository)
- Documentation
- Architecture diagram
- Final presentation

3. Tools and Technologies

Languages

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- SQL
- Python

Big Data

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- Apache Spark (PySpark)
- Apache Kafka

Cloud

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- Microsoft Azure

Orchestration

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- Apache Airflow
- Azure Data Factory

Databases

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- PostgreSQL
- MySQL
- MongoDB

4. Final Guidance

- Focus on hands-on practice
 - Build real-world data pipelines
 - Keep concepts clear and structured
 - Practice debugging and optimization
 - Maintain proper documentation for all work
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